

Contents

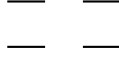
1	Orbitals And Their Symmetry	2
2	Linear Combinations: 2 Monomer- / 1 Dimer-States	3
3	Linear Combinations: 4 Monomer- / 2 Dimer-States	21
4	Linear Combinations: 8 Monomer- / 4 Dimer-States	32
5	Linear Combinations: 16 Monomer- / 8 Dimer-States	48

1 Orbitals And Their Symmetry

We use the following order of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_2^* & \text{---} & \text{---} & a_2^* \\ a_2 & \text{---} & \text{---} & b_2 \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:



where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital order within the group of monomer orbitals follows the above mentioned definition. The transformation of

monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$a_2^{*l} = +a_u^* + b_g^* \tag{1a}$$

$$b_2^{*l} = +a_g^* + b_u^* \tag{1b}$$

$$a_2^l = +a_u + b_g \tag{1c}$$

$$b_2^l = +a_g + b_u \tag{1d}$$

$$a_2^{*r} = +a_u^* - b_g^* \tag{1e}$$

$$b_2^{*r} = -a_g^* + b_u^* \tag{1f}$$

$$a_2^r = +a_u - b_g \tag{1g}$$

$$b_2^r = -a_g + b_u \tag{1h}$$

2 Linear Combinations: 2 Monomer- / 1 Dimer-States

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & \uparrow \\ \uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & - \\ \downarrow\downarrow & \downarrow\downarrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^{*l}) \cdot (a_2^l) \cdot (b_2^l)}_{l: a_1} \Rightarrow + |(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*l})| \\
& = + |(+a_g + b_u) \cdot (+a_g^* + b_u^*) \cdot (+a_u + b_g) \cdot (+a_u^* + b_g^*)| \\
& = + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* a_u b_g^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
& \quad + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_u^* a_u b_g^*}_{a_g} \right| + \left| \underbrace{a_g b_u^* b_g a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_u^* b_g b_g^*}_{b_u} \right| \\
& \quad + \left| \underbrace{b_u a_g^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{b_u a_g^* a_u b_g^*}_{a_g} \right| + \left| \underbrace{b_u a_g^* b_g a_u^*}_{a_g} \right| + \left| \underbrace{b_u a_g^* b_g b_g^*}_{b_u} \right| \\
& \quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_u b_u^* a_u b_g^*}_{b_u} \right| + \left| \underbrace{b_u b_u^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{b_u b_u^* b_g b_g^*}_{a_g} \right| \\
& = + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
& \quad + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\
& \quad - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\
& \quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc} - & - \\ \downarrow\downarrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} \uparrow & \uparrow \\ \uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*r}) \cdot (a_2^{*r}) \cdot (a_2^r) \cdot (b_2^r)}_{r: a_1} \Rightarrow + |(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*r})| \\
& = + |(-a_g + b_u) \cdot (-a_g^* + b_u^*) \cdot (+a_u - b_g) \cdot (+a_u^* - b_g^*)|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* a_u b_g^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
& - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_u^* a_u b_g^*}_{a_g} \right| + \left| \underbrace{a_g b_u^* b_g a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_u^* b_g b_g^*}_{b_u} \right| \\
& - \left| \underbrace{b_u a_g^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{b_u a_g^* a_u b_g^*}_{a_g} \right| + \left| \underbrace{b_u a_g^* b_g a_u^*}_{a_g} \right| - \left| \underbrace{b_u a_g^* b_g b_g^*}_{b_u} \right| \\
& + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_u b_u^* a_u b_g^*}_{b_u} \right| - \left| \underbrace{b_u b_u^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{b_u b_u^* b_g b_g^*}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
& - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\
& + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\
& + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \end{array} \right) \\
& + \underbrace{(a_2^{*l}) \cdot (a_2^l)}_{\text{l: } a_1} \cdot \underbrace{(a_2^{*r}) \cdot (a_2^r)}_{\text{r: } a_1} \Rightarrow + |(a_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(+a_u + b_g) \cdot (+a_u - b_g) \cdot (+a_u^* + b_g^*) \cdot (+a_u^* - b_g^*)| \\
& = + \left| \underbrace{a_u b_g a_u^* b_g^*}_{a_g} \right| - \left| \underbrace{a_u b_g b_g^* a_u^*}_{a_g} \right| - \left| \underbrace{b_g a_u a_u^* b_g^*}_{a_g} \right| + \left| \underbrace{b_g a_u b_g^* a_u^*}_{a_g} \right| \\
& = - \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(a_2^{*l}) \cdot (a_2^l)}_{l: a_1} \cdot \underbrace{(b_2^{*r}) \cdot (a_2^r)}_{r: b_1} \Rightarrow + |(b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*l})| \\
& = + |(-a_g^* + b_u^*) \cdot (+a_u + b_g) \cdot (+a_u - b_g) \cdot (+a_u^* + b_g^*)| \\
\\
& = + \left| \underbrace{a_g^* a_u b_g a_u^*}_{b_g} \right| + \left| \underbrace{a_g^* a_u b_g b_g^*}_{a_u} \right| - \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g^* b_g a_u b_g^*}_{a_u} \right| \\
& \quad - \left| \underbrace{b_u^* a_u b_g a_u^*}_{a_u} \right| - \left| \underbrace{b_u^* a_u b_g b_g^*}_{b_g} \right| + \left| \underbrace{b_u^* b_g a_u a_u^*}_{a_u} \right| + \left| \underbrace{b_u^* b_g a_u b_g^*}_{b_g} \right| \\
\\
& = - \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| \\
& \quad - \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| - \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^l)}_{l: b_1} \cdot \underbrace{(a_2^{*r}) \cdot (a_2^r)}_{r: a_1} \Rightarrow + |(b_2^{*l}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*r})| \\
& = + |(+a_g^* + b_u^*) \cdot (+a_u + b_g) \cdot (+a_u - b_g) \cdot (+a_u^* - b_g^*)| \\
\\
& = - \left| \underbrace{a_g^* a_u b_g a_u^*}_{b_g} \right| + \left| \underbrace{a_g^* a_u b_g b_g^*}_{a_u} \right| + \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g^* b_g a_u b_g^*}_{a_u} \right| \\
& \quad - \left| \underbrace{b_u^* a_u b_g a_u^*}_{a_u} \right| + \left| \underbrace{b_u^* a_u b_g b_g^*}_{b_g} \right| + \left| \underbrace{b_u^* b_g a_u a_u^*}_{a_u} \right| - \left| \underbrace{b_u^* b_g a_u b_g^*}_{b_g} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| \\
& - \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| - \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right|
\end{aligned}$$

summarized:

$$= +2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow & \uparrow \end{array} \right) \\
& + \underbrace{(a_2^{*l}) \cdot (a_2^l)}_{\text{l: } a_1} \cdot \underbrace{(a_2^{*r}) \cdot (b_2^r)}_{\text{r: } b_1} \Rightarrow + |(b_2^r) \cdot (a_2^l) \cdot (a_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(-a_g + b_u) \cdot (+a_u + b_g) \cdot (+a_u^* + b_g^*) \cdot (+a_u^* - b_g^*)| \\
\\
& = + \left| \underbrace{a_g a_u a_u^* b_g^*}_{b_g} \right| - \left| \underbrace{a_g a_u b_g^* a_u^*}_{b_g} \right| + \left| \underbrace{a_g b_g a_u^* b_g^*}_{a_u} \right| - \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| \\
& \quad - \left| \underbrace{b_u a_u a_u^* b_g^*}_{a_u} \right| + \left| \underbrace{b_u a_u b_g^* a_u^*}_{a_u} \right| - \left| \underbrace{b_u b_g a_u^* b_g^*}_{b_g} \right| + \left| \underbrace{b_u b_g b_g^* a_u^*}_{b_g} \right| \\
\\
& = + \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| \\
& \quad + \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| + \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

summarized:

$$= + 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc} - & \uparrow \\ \downarrow \downarrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \downarrow \end{array} \right) \\
& + \underbrace{(a_2^{*l}) \cdot (b_2^l)}_{\text{l: } b_1} \cdot \underbrace{(a_2^{*r}) \cdot (a_2^r)}_{\text{r: } a_1} \Rightarrow + |(b_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(+a_g + b_u) \cdot (+a_u - b_g) \cdot (+a_u^* + b_g^*) \cdot (+a_u^* - b_g^*)| \\
\\
& = - \left| \underbrace{a_g a_u a_u^* b_g^*}_{b_g} \right| + \left| \underbrace{a_g a_u b_g^* a_u^*}_{b_g} \right| + \left| \underbrace{a_g b_g a_u^* b_g^*}_{a_u} \right| - \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| \\
& \quad - \left| \underbrace{b_u a_u a_u^* b_g^*}_{a_u} \right| + \left| \underbrace{b_u a_u b_g^* a_u^*}_{a_u} \right| + \left| \underbrace{b_u b_g a_u^* b_g^*}_{b_g} \right| - \left| \underbrace{b_u b_g b_g^* a_u^*}_{b_g} \right|
\end{aligned}$$

$$\begin{aligned}
= & - \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| \\
& + \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| - \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \updownarrow & \updownarrow & \uparrow \end{pmatrix} \\
& + \underbrace{(a_2^{*l}) \cdot (a_2^l)}_{\text{l: } a_1} \cdot \underbrace{(b_2^{*r}) \cdot (b_2^r)}_{\text{r: } a_1} \Rightarrow + |(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^{*l})| \\
& = + |(-a_g + b_u) \cdot (-a_g^* + b_u^*) \cdot (+a_u + b_g) \cdot (+a_u^* + b_g^*)| \\
& = + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* a_u b_g^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
& \quad - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_u^* a_u b_g^*}_{a_g} \right| - \left| \underbrace{a_g b_u^* b_g a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_u^* b_g b_g^*}_{b_u} \right| \\
& \quad - \left| \underbrace{b_u a_g^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{b_u a_g^* a_u b_g^*}_{a_g} \right| - \left| \underbrace{b_u a_g^* b_g a_u^*}_{a_g} \right| - \left| \underbrace{b_u a_g^* b_g b_g^*}_{b_u} \right| \\
& \quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_u b_u^* a_u b_g^*}_{b_u} \right| + \left| \underbrace{b_u b_u^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{b_u b_u^* b_g b_g^*}_{a_g} \right| \\
& = + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
& \quad - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\
& \quad + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\
& \quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \updownarrow & \uparrow & \uparrow & \updownarrow \end{pmatrix} \\
& + \underbrace{(b_2^{*l}) \cdot (b_2^l)}_{\text{l: } a_1} \cdot \underbrace{(a_2^{*r}) \cdot (a_2^r)}_{\text{r: } a_1} \Rightarrow + |(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^r) \cdot (a_2^{*r})| \\
& = + |(+a_g + b_u) \cdot (+a_g^* + b_u^*) \cdot (+a_u - b_g) \cdot (+a_u^* - b_g^*)|
\end{aligned}$$

$$\begin{aligned}
&= + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* a_u b_g^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
&\quad + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_u^* a_u b_g^*}_{a_g} \right| - \left| \underbrace{a_g b_u^* b_g a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_u^* b_g b_g^*}_{b_u} \right| \\
&\quad + \left| \underbrace{b_u a_g^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{b_u a_g^* a_u b_g^*}_{a_g} \right| - \left| \underbrace{b_u a_g^* b_g a_u^*}_{a_g} \right| + \left| \underbrace{b_u a_g^* b_g b_g^*}_{b_u} \right| \\
&\quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_u b_u^* a_u b_g^*}_{b_u} \right| - \left| \underbrace{b_u b_u^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{b_u b_u^* b_g b_g^*}_{a_g} \right| \\
&= + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
&\quad + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\
&\quad - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\
&\quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow \uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^l)}_{\text{l: } b_1} \cdot \underbrace{(b_2^{*r}) \cdot (a_2^r)}_{\text{r: } b_1} \Rightarrow + |(b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r)| \\
& = + |(+ a_g^* + b_u^*) \cdot (-a_g^* + b_u^*) \cdot (+a_u + b_g) \cdot (+a_u - b_g)| \\
\\
& = - \left| \underbrace{a_g^* b_u^* a_u b_g}_{a_g} \right| + \left| \underbrace{a_g^* b_u^* b_g a_u}_{a_g} \right| + \left| \underbrace{b_u^* a_g^* a_u b_g}_{a_g} \right| - \left| \underbrace{b_u^* a_g^* b_g a_u}_{a_g} \right| \\
\\
& = - \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} - & \\ \downarrow\uparrow & \downarrow\uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^l)}_{\text{l: } b_1} \cdot \underbrace{(a_2^{*r}) \cdot (b_2^r)}_{\text{r: } b_1} \Rightarrow + |(b_2^r) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*r})| \\
& = + |(-a_g + b_u) \cdot (+a_g^* + b_u^*) \cdot (+a_u + b_g) \cdot (+a_u^* - b_g^*)| \\
\\
& = - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* a_u b_g^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
& \quad - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_u^* a_u b_g^*}_{a_g} \right| - \left| \underbrace{a_g b_u^* b_g a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_u^* b_g b_g^*}_{b_u} \right| \\
& \quad + \left| \underbrace{b_u a_g^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{b_u a_g^* a_u b_g^*}_{a_g} \right| + \left| \underbrace{b_u a_g^* b_g a_u^*}_{a_g} \right| - \left| \underbrace{b_u a_g^* b_g b_g^*}_{b_u} \right| \\
& \quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_u b_u^* a_u b_g^*}_{b_u} \right| + \left| \underbrace{b_u b_u^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{b_u b_u^* b_g b_g^*}_{a_g} \right| \\
\\
& = - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
& \quad - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\
& \quad - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\
& \quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc} - & \uparrow \\ \downarrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right) \\
& + \underbrace{(a_2^{*l}) \cdot (b_2^l)}_{\text{l: } b_1} \cdot \underbrace{(b_2^{*r}) \cdot (a_2^r)}_{\text{r: } b_1} \Rightarrow + |(b_2^l) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*l})| \\
& = + |(+a_g + b_u) \cdot (-a_g^* + b_u^*) \cdot (+a_u - b_g) \cdot (+a_u^* + b_g^*)|
\end{aligned}$$

$$\begin{aligned}
&= - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* a_u b_g^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
&\quad + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_u^* a_u b_g^*}_{a_g} \right| - \left| \underbrace{a_g b_u^* b_g a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_u^* b_g b_g^*}_{b_u} \right| \\
&\quad - \left| \underbrace{b_u a_g^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{b_u a_g^* a_u b_g^*}_{a_g} \right| + \left| \underbrace{b_u a_g^* b_g a_u^*}_{a_g} \right| + \left| \underbrace{b_u a_g^* b_g b_g^*}_{b_u} \right| \\
&\quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_u b_u^* a_u b_g^*}_{b_u} \right| - \left| \underbrace{b_u b_u^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{b_u b_u^* b_g b_g^*}_{a_g} \right| \\
&= - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\
&\quad + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\
&\quad + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\
&\quad + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^l)}_{l: b_1} \cdot \underbrace{(b_2^{*r}) \cdot (b_2^r)}_{r: a_1} \Rightarrow + |(b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l)| \\
& = + |(-a_g + b_u) \cdot (+a_g^* + b_u^*) \cdot (-a_g^* + b_u^*) \cdot (+a_u + b_g)| \\
& = - \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| - \left| \underbrace{a_g a_g^* b_u^* b_g}_{a_u} \right| + \left| \underbrace{a_g b_u^* a_g^* a_u}_{b_g} \right| + \left| \underbrace{a_g b_u^* a_g^* b_g}_{a_u} \right| \\
& \quad + \left| \underbrace{b_u a_g^* b_u^* a_u}_{a_u} \right| + \left| \underbrace{b_u a_g^* b_u^* b_g}_{b_g} \right| - \left| \underbrace{b_u b_u^* a_g^* a_u}_{a_u} \right| - \left| \underbrace{b_u b_u^* a_g^* b_g}_{b_g} \right| \\
& = - \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| \\
& \quad - \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| - \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (b_2^l)}_{l: a_1} \cdot \underbrace{(b_2^{*r}) \cdot (a_2^r)}_{r: b_1} \Rightarrow + |(b_2^l) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^r)| \\
& = + |(+a_g + b_u) \cdot (+a_g^* + b_u^*) \cdot (-a_g^* + b_u^*) \cdot (+a_u - b_g)| \\
& = + \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| - \left| \underbrace{a_g a_g^* b_u^* b_g}_{a_u} \right| - \left| \underbrace{a_g b_u^* a_g^* a_u}_{b_g} \right| + \left| \underbrace{a_g b_u^* a_g^* b_g}_{a_u} \right| \\
& \quad + \left| \underbrace{b_u a_g^* b_u^* a_u}_{a_u} \right| - \left| \underbrace{b_u a_g^* b_u^* b_g}_{b_g} \right| - \left| \underbrace{b_u b_u^* a_g^* a_u}_{a_u} \right| + \left| \underbrace{b_u b_u^* a_g^* b_g}_{b_g} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| + \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| \\
& - \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| - \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right|
\end{aligned}$$

summarized:

$$= +2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc} - & \uparrow \\ \downarrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow\uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(a_2^{*l}) \cdot (b_2^l)}_{\text{l: } b_1} \cdot \underbrace{(a_2^{*r}) \cdot (b_2^r)}_{\text{r: } b_1} \Rightarrow + |(b_2^l) \cdot (b_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(+a_g + b_u) \cdot (-a_g + b_u) \cdot (+a_u^* + b_g^*) \cdot (+a_u^* - b_g^*)| \\
\\
& = - \left| \underbrace{a_g b_u a_u^* b_g^*}_{a_g} \right| + \left| \underbrace{a_g b_u b_g^* a_u^*}_{a_g} \right| + \left| \underbrace{b_u a_g a_u^* b_g^*}_{a_g} \right| - \left| \underbrace{b_u a_g b_g^* a_u^*}_{a_g} \right| \\
\\
& = - \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(a_2^{*l}) \cdot (b_2^l)}_{\text{l: } b_1} \cdot \underbrace{(b_2^{*r}) \cdot (b_2^r)}_{\text{r: } a_1} \Rightarrow + |(b_2^l) \cdot (b_2^r) \cdot (b_2^{*r}) \cdot (a_2^{*l})| \\
& = + |(a_g + b_u) \cdot (-a_g + b_u) \cdot (-a_g^* + b_u^*) \cdot (+a_u^* + b_g^*)| \\
& = - \left| \underbrace{a_g b_u a_g^* a_u^*}_{b_g} \right| - \left| \underbrace{a_g b_u a_g^* b_g^*}_{a_u} \right| + \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + \left| \underbrace{a_g b_u b_u^* b_g^*}_{b_g} \right| \\
& \quad + \left| \underbrace{b_u a_g a_g^* a_u^*}_{b_g} \right| + \left| \underbrace{b_u a_g a_g^* b_g^*}_{a_u} \right| - \left| \underbrace{b_u a_g b_u^* a_u^*}_{a_u} \right| - \left| \underbrace{b_u a_g b_u^* b_g^*}_{b_g} \right| \\
& = + \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& \quad + \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& = + 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& \quad + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& \quad + \underbrace{(b_2^{*l}) \cdot (b_2^l)}_{\text{l: } a_1} \cdot \underbrace{(a_2^{*r}) \cdot (b_2^r)}_{\text{r: } b_1} \Rightarrow + |(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(a_g + b_u) \cdot (-a_g + b_u) \cdot (+a_g^* + b_u^*) \cdot (+a_u^* - b_g^*)| \\
& = + \left| \underbrace{a_g b_u a_g^* a_u^*}_{b_g} \right| - \left| \underbrace{a_g b_u a_g^* b_g^*}_{a_u} \right| + \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - \left| \underbrace{a_g b_u b_u^* b_g^*}_{b_g} \right| \\
& \quad - \left| \underbrace{b_u a_g a_g^* a_u^*}_{b_g} \right| + \left| \underbrace{b_u a_g a_g^* b_g^*}_{a_u} \right| - \left| \underbrace{b_u a_g b_u^* a_u^*}_{a_u} \right| + \left| \underbrace{b_u a_g b_u^* b_g^*}_{b_g} \right|
\end{aligned}$$

$$\begin{aligned}
= & - \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& - \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (b_2^l)}_{\text{l: } a_1} \cdot \underbrace{(b_2^{*r}) \cdot (b_2^r)}_{\text{r: } a_1} \Rightarrow + |(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r})| \\
& = + |(+a_g + b_u) \cdot (-a_g + b_u) \cdot (+a_g^* + b_u^*) \cdot (-a_g^* + b_u^*)| \\
& = + \left| \underbrace{a_g b_u a_g^* b_u^*}_{a_g} \right| - \left| \underbrace{a_g b_u b_u^* a_g^*}_{a_g} \right| - \left| \underbrace{b_u a_g a_g^* b_u^*}_{a_g} \right| + \left| \underbrace{b_u a_g b_u^* a_g^*}_{a_g} \right| \\
& = - \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right|$$

3 Linear Combinations: 4 Monomer- / 2 Dimer-States

$$+ \begin{pmatrix} \uparrow & \uparrow & - & - \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & - & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_g, a_1

$$= + (|(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*l})|) + (|(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= +2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| + 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| \\ - 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|$$

$$+ \begin{pmatrix} \uparrow & \uparrow & - & - \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & - & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_g, a_1

$$= + (|(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*l})|) - (|(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= -2 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + 2 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - 2 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| - 2 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + 2 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right|$$

$$+\left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array}\right)+\left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array}\right)$$

$$\text{enthaltene Monomer-Symmetrien} = a_1$$

$$= \qquad \qquad \qquad + (|(a_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|) \qquad \qquad + \qquad \qquad (|(a_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad -8 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= + (|(b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*l})|) + (|(b_2^{*l}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= +4 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 4 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= + (|(b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*l})|) - (|(b_2^{*l}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= -4 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 4 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \text{---} & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \text{---} & \uparrow & \text{---} & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= + (|(b_2^r) \cdot (a_2^l) \cdot (a_2^{*l}) \cdot (a_2^{*r})|) + (|(b_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|)$$

$$= -4 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 4 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \text{---} & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \text{---} & \uparrow & \text{---} & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= + (|(b_2^r) \cdot (a_2^l) \cdot (a_2^{*l}) \cdot (a_2^{*r})|) - (|(b_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|)$$

$$= +4 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 4 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \uparrow & \updownarrow & \updownarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \updownarrow & \uparrow & \uparrow & \updownarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1

$$= \quad + (|(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^{*l})|) \quad + \quad (|(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= +2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| - 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| \\ + 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \uparrow & \updownarrow & \updownarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \updownarrow & \uparrow & \uparrow & \updownarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1

$$= \quad + (|(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^{*l})|) \quad - \quad (|(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= -2 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + 2 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + 2 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| - 2 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + 2 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right|$$

$$+\left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array}\right)+\left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array}\right)$$

enthaltene Monomer-Symmetrien = b_1

$$= \qquad \qquad \qquad +\left(|(b_2^{*l})\cdot (b_2^{*r})\cdot (a_2^l)\cdot (a_2^r)|\right) \qquad \qquad + \qquad \qquad \qquad \left(|(b_2^{*l})\cdot (b_2^{*r})\cdot (a_2^l)\cdot (a_2^r)|\right)$$

$$= \qquad \qquad \qquad -8\left|\underbrace{a_g^*b_gb_u^*}_{a_g}a_u\right|$$

$$+ \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_1

$$= + (|(b_2^r) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*r})|) + (|(b_2^l) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*l})|)$$

$$= -2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| + 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| \\ + 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right|$$

$$+ \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_1

$$= + (|(b_2^r) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*r})|) - (|(b_2^l) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*l})|)$$

$$= -2 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - 2 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - 2 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| + 2 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + 2 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right|$$

$$+ \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \quad + (|(b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l)|) \quad + \quad (|(b_2^l) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^r)|)$$

$$= \quad + 4 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| \quad - \quad 4 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right|$$

$$+ \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \quad + (|(b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l)|) \quad - \quad (|(b_2^l) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^r)|)$$

$$= \quad - 4 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| \quad - \quad 4 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right|$$

$$+ \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_1

$$= \qquad \qquad \qquad + (|(b_2^l) \cdot (b_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|) \qquad \qquad + \qquad \qquad (|(b_2^l) \cdot (b_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad -8 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \qquad \qquad \qquad + (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*r}) \cdot (a_2^{*l})|) \qquad \qquad + \qquad \qquad (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad -4 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| \qquad \qquad + \qquad \qquad 4 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \qquad \qquad \qquad + (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*r}) \cdot (a_2^{*l})|) \qquad \qquad - \qquad \qquad (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad +4 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| \qquad \qquad + \qquad \qquad 4 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right|$$

$$+ \left(\begin{array}{cc} \uparrow & - \\ \downarrow \uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \downarrow \uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow \uparrow & \uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = a_1

$$= \qquad \qquad \qquad + (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r})|) \qquad \qquad \qquad + \qquad \qquad \qquad (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r})|)$$

$$= \qquad \qquad \qquad -8 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right|$$

4 Linear Combinations: 8 Monomer- / 4 Dimer-States

$i^3 a_1 * i^3 a_1$:

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| + 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| - 4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|
\end{aligned}$$

$$i^3 a_1 * e^3 a_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - 2 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| \\
& + 2 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| + 2 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| \\
& - 2 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + 4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|
\end{aligned}$$

$$i^3 a_1 * e^3 b_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& = +2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\
& + 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& - 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& - 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

$$i^3 a_1 * i^3 b_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& = +2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\
& \quad + 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad + 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& \quad + 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

$$e^3 a_1 * i^3 a_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + 2 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| \\
& \quad - 2 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| - 2 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| \\
& \quad + 2 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + 4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|
\end{aligned}$$

$$e^3 a_1 * e^3 a_1:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| \right) + \left(\begin{array}{c} + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} - \underbrace{a_g a_g^* b_g^* a_u}_{b_u} + \underbrace{a_g a_g^* b_g a_u^*}_{b_u} + \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} - \underbrace{a_g b_g^* b_u^* a_u}_{a_g} + \underbrace{a_g b_g b_u^* a_u^*}_{a_g} - \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} + \underbrace{a_g^* b_g^* b_u a_u}_{a_g} - \underbrace{a_g^* b_g b_u a_u^*}_{a_g} + \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} - \underbrace{b_g^* b_u b_u^* a_u}_{b_u} + \underbrace{b_g b_u b_u^* a_u^*}_{b_u} + \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} + \underbrace{a_g a_g^* b_g^* a_u}_{b_u} - \underbrace{a_g a_g^* b_g a_u^*}_{b_u} + \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} - \underbrace{a_g b_g^* b_u^* a_u}_{a_g} + \underbrace{a_g b_g b_u^* a_u^*}_{a_g} + \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} + \underbrace{a_g^* b_g^* b_u a_u}_{a_g} - \underbrace{a_g^* b_g b_u a_u^*}_{a_g} - \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} + \underbrace{b_g^* b_u b_u^* a_u}_{b_u} - \underbrace{b_g b_u b_u^* a_u^*}_{b_u} + \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) + \left(-4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| - 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| \\ + 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| \\ + 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| - 4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|$$

$$e^3 a_1 * e^3 b_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& = +2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\
& - 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& - 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& + 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

$$e^3 a_1 * i^3 b_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& = +2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\
& \quad - 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad + 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& \quad - 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

$$e^3 b_1 * i^3 a_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& = -2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\
& + 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& - 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& + 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

$$e^3 b_1 * e^3 a_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& + \left(-2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& = -2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\
& \quad + 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& \quad + 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad - 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

$$e^3 b_1 * e^3 b_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| \right) \\
& + \left(\begin{array}{c} - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| + 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| \\
& + 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| \\
& - 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| - 4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|
\end{aligned}$$

$$e^3 b_1 * i^3 b_1:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| \right) + \left(\begin{array}{c} - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) - \left(+4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + 2 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| \\ - 2 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| - 2 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| \\ - 2 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + 4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|$$

$$i^3 b_1 * i^3 a_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& = -2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\
& \quad - 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& \quad - 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad - 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

$$i^3 b_1 * e^3 a_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& + \left(-2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
& = -2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\
& \quad - 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\
& \quad + 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad + 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

$$i^3 b_1 * e^3 b_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& + \left(+4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - 2 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| \\
& + 2 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| - 2 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| + 2 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| \\
& + 2 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + 4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|
\end{aligned}$$

$$i^3 b_1 * i^3 b_1:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| \right) - \left(\begin{array}{c} + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) + \left(-4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| \right)$$

$$= -4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| - 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| \\ - 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| \\ + 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| - 4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|$$

5 Linear Combinations: 16 Monomer- / 8 Dimer-States

identical $i^3 a_1$ and $i^3 a_1$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| + 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| - 4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|
\end{aligned}$$

plus combination: i^3a_1 and $e^3a_1 + e^3a_1$ and i^3a_1

$$\begin{aligned}
 &+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) \\
 &+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{array} \right)
 \end{aligned}$$

[illegible]

$$= -8 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| + 8 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|$$

minus combination: i^3a_1 and $e^3a_1 - e^3a_1$ and i^3a_1

$$\begin{aligned} &+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \\ &- \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \end{aligned}$$

[illegible]

$$\begin{aligned}
&= +4 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - 4 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + 4 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + 4 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\
&\qquad\qquad\qquad - 4 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - 4 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| + 4 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - 4 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right|
\end{aligned}$$

plus combination: $i^3 a_1$ and $e^3 b_1 + e^3 b_1$ and $i^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 4 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 4 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 4 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| \\
& \quad - 4 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 4 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 4 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 4 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right|
\end{aligned}$$

minus combination: $i^3 a_1$ and $e^3 b_1 - e^3 b_1$ and $i^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 4 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| + 4 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 4 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad - 4 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 4 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| - 4 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 4 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

plus combination: $i^3 a_1$ and $i^3 b_1 + i^3 b_1$ and $i^3 a_1$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

$$+ \begin{pmatrix} +2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\ - \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\ - \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\ + \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\ + \left(-2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\ - \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\ - \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\ + \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \end{pmatrix}$$

$$= +4 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 4 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 4 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 4 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| \\ + 4 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 4 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 4 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 4 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right|$$

minus combination: $i^3 a_1$ and $i^3 b_1$ - $i^3 b_1$ and $i^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* l_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g l_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* l_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g l_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 4 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| + 4 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 4 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad + 4 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 4 \left| \underbrace{a_g l_g^* b_u b_u^*}_{b_g} \right| + 4 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 4 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

identical $e^3 a_1$ and $e^3 a_1$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow\uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} - \underbrace{a_g a_g^* b_g^* a_u}_{b_u} + \underbrace{a_g a_g^* b_g a_u^*}_{b_u} + \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} - \underbrace{a_g b_g^* b_u^* a_u}_{a_g} + \underbrace{a_g b_g b_u^* a_u^*}_{a_g} - \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} + \underbrace{a_g^* b_g^* b_u a_u}_{a_g} - \underbrace{a_g^* b_g b_u a_u^*}_{a_g} + \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} - \underbrace{b_g^* b_u b_u^* a_u}_{b_u} + \underbrace{b_g b_u b_u^* a_u^*}_{b_u} + \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} + \underbrace{a_g a_g^* b_g^* a_u}_{b_u} - \underbrace{a_g a_g^* b_g a_u^*}_{b_u} + \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} - \underbrace{a_g b_g^* b_u^* a_u}_{a_g} + \underbrace{a_g b_g b_u^* a_u^*}_{a_g} + \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} + \underbrace{a_g^* b_g^* b_u a_u}_{a_g} - \underbrace{a_g^* b_g b_u a_u^*}_{a_g} - \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} + \underbrace{b_g^* b_u b_u^* a_u}_{b_u} - \underbrace{b_g b_u b_u^* a_u^*}_{b_u} + \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g a_g^* b_u b_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| - 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| \\
& + 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| \\
& + 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| - 4 \left| \underbrace{b_g b_g^* a_u a_u^*}_{a_g} \right|
\end{aligned}$$

plus combination: $e^3 a_1$ and $e^3 b_1 + e^3 b_1$ and $e^3 a_1$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

$$+ \begin{pmatrix} +2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\ - 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\ - 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\ + 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \\ - 2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \\ + 2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \\ + 2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\ - 2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \end{pmatrix}$$

$$= +4 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 4 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 4 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 4 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| \\ - 4 \left| \underbrace{a_g a_g^* b_g b_u}_{a_u} \right| + 4 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 4 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 4 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right|$$

minus combination: $e^3 a_1$ and $e^3 b_1 - e^3 b_1$ and $e^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 4 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| - 4 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 4 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad - 4 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 4 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| + 4 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 4 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

plus combination: $e^3 a_1$ and $i^3 b_1 + i^3 b_1$ and $e^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g b_g^* b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 4 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 4 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 4 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| \\
& \quad + 4 \left| \underbrace{a_g a_g^* b_g^* b_u}_{a_u} \right| - 4 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 4 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 4 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right|
\end{aligned}$$

minus combination: $e^3 a_1$ and $i^3 b_1 - i^3 b_1$ and $e^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| - 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| - 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g a_g^* l_g^* b_u}_{a_u} \right| - 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g l_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| + 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right) \\
& - \left(+2 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g b_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_u b_u^* a_u}_{a_u} \right| + 2 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| \right) \\
& + \left(+2 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g a_g^* l_g^* b_u}_{a_u} \right| + 2 \left| \underbrace{a_g b_u b_u^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{a_g l_g^* b_u b_u^*}_{b_g} \right| \right) \\
& - \left(-2 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g^* b_g b_g^* a_u}_{a_u} \right| + 2 \left| \underbrace{b_g b_u^* a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \right) \\
& + \left(-2 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 2 \left| \underbrace{a_g b_g b_g^* a_u^*}_{a_u} \right| + 2 \left| \underbrace{b_g^* b_u a_u a_u^*}_{a_u} \right| - 2 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right| \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_g a_g^* b_u^* a_u}_{b_g} \right| + 4 \left| \underbrace{a_g^* b_g b_u b_u^*}_{b_g} \right| - 4 \left| \underbrace{a_g^* b_g a_u a_u^*}_{b_g} \right| - 4 \left| \underbrace{b_g b_g^* b_u^* a_u}_{b_g} \right| \\
& \quad + 4 \left| \underbrace{a_g a_g^* b_u a_u^*}_{b_g} \right| + 4 \left| \underbrace{a_g l_g^* b_u b_u^*}_{b_g} \right| - 4 \left| \underbrace{a_g b_g^* a_u a_u^*}_{b_g} \right| - 4 \left| \underbrace{b_g b_g^* b_u a_u^*}_{b_g} \right|
\end{aligned}$$

identical e^3b_1 and e^3b_1

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| \right) \\
& + \left(\begin{array}{c} - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ - \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| + \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| + \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ + \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| - \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| + 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| \\
& + 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| + 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| \\
& - 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| - 4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|
\end{aligned}$$

plus combination: e^3b_1 and $i^3b_1 + i^3b_1$ and e^3b_1

$$\begin{aligned}
 &+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
 &+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
 \end{aligned}$$

[illegible]

$$= -8 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| + 8 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|$$

minus combination: e^3b_1 and i^3b_1 - i^3b_1 and e^3b_1

$$\begin{aligned}
 &+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
 &- \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
 \end{aligned}$$

[illegible]

$$\begin{aligned}
&= +4 \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + 4 \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| + 4 \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - 4 \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\
&\qquad\qquad\qquad + 4 \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - 4 \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| - 4 \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - 4 \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right|
\end{aligned}$$

identical $i^3 b_1$ and $i^3 b_1$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| \right) \\
& - \left(\begin{array}{c} + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ - \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| + \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ - \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| + \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \\ + \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| + \left| \underbrace{a_g a_g^* b_g^* a_u}_{b_u} \right| + \left| \underbrace{a_g a_g^* b_g a_u^*}_{b_u} \right| - \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| \\ + \left| \underbrace{a_g b_u^* a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| - \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - \left| \underbrace{a_g b_g b_g^* b_u^*}_{b_u} \right| \\ + \left| \underbrace{a_g^* b_u a_u a_u^*}_{b_u} \right| - \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - \left| \underbrace{a_g^* b_g b_g^* b_u}_{b_u} \right| \\ - \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| - \left| \underbrace{b_g^* b_u b_u^* a_u}_{b_u} \right| - \left| \underbrace{b_g b_u b_u^* a_u^*}_{b_u} \right| + \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right| \right) \\
& = -4 \left| \underbrace{a_g^* b_g b_u^* a_u}_{a_g} \right| + 2 \left| \underbrace{a_g a_g^* a_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g a_g^* b_g b_g^*}_{a_g} \right| - 2 \left| \underbrace{a_g b_g^* b_u^* a_u}_{a_g} \right| \\
& \quad - 2 \left| \underbrace{a_g b_g b_u^* a_u^*}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g^* b_u a_u}_{a_g} \right| - 2 \left| \underbrace{a_g^* b_g b_u a_u^*}_{a_g} \right| - 2 \left| \underbrace{b_u b_u^* a_u a_u^*}_{a_g} \right| \\
& \quad + 2 \left| \underbrace{b_g b_g^* b_u b_u^*}_{a_g} \right| - 4 \left| \underbrace{a_g b_g^* b_u a_u^*}_{a_g} \right|
\end{aligned}$$